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2. Introduction

SPSS (Statistical Package for the Social Sciences) is a widely used statistical software designed for data analysis in social sciences, business, health research, education, and various other fields. Originally developed by three Stanford University graduate students in the late 1960s, SPSS became popular for enabling researchers to perform complex statistical analyses without requiring advanced programming skills. In 2009, IBM acquired SPSS and rebranded it as **IBM SPSS Statistics**, though most users still refer to it simply as SPSS due to its longstanding recognition.

SPSS offers a powerful combination of a user-friendly graphical interface and a command syntax language, allowing both beginners and professional statisticians to manage data, perform statistical tests, and create detailed reports. It simplifies data manipulation by organizing datasets in a two-dimensional table where rows represent cases (such as individuals) and columns represent variables (such as age or income). SPSS supports only two basic data types: numeric and string (text), which streamlines data processing. The software includes features for data cleaning, statistical modeling (like ANOVA, regression), and advanced reporting, making it a comprehensive tool for both simple and complex analytical tasks.

Additionally, SPSS integrates with scripting languages like Python, enabling automation of repetitive tasks and extending its capabilities to interact with external data sources. The output generated by SPSS can be exported into formats like Word, Excel, PDF, and various image formats, facilitating easy sharing and documentation. Whether used by researchers, students, or analysts, SPSS remains a preferred tool for statistical analysis due to its accessibility, flexibility, and robust analytical power.

3. Scope of SPSS

1. Data Visualization

SPSS provides powerful tools for creating visual representations of data such as charts, histograms, scatter plots, and box plots. These visuals help researchers and analysts to better understand patterns, trends, and outliers in the dataset, making data interpretation more intuitive and accessible.

2. Advanced Data Analysis

SPSS is widely used for performing complex statistical analyses like regression, factor analysis, cluster analysis, and multivariate analysis. These advanced techniques help in uncovering hidden relationships between variables and in making precise data-driven predictions.

3. Behavioral Analysis

Psychologists, sociologists, and market analysts use SPSS to study and interpret human behavior patterns. Through the application of statistical tests on survey and observational data, SPSS aids in understanding behaviors, preferences, and psychological trends in various populations.

4. Predictive Analytics

SPSS is a strong tool for predictive modeling, enabling organizations to forecast future outcomes based on historical data. It is commonly used in fields like finance, marketing, and healthcare for demand forecasting, risk analysis, and predicting customer churn.

5. Data Management & Cleaning

SPSS simplifies data management by offering features like data validation, transformation, and cleaning. It helps in handling missing values, recoding variables, and merging datasets efficiently, ensuring high-quality data for accurate analysis.

4. Show the following data in Bar Graph and Pie Chart.

Variable view

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
Student	String	12	0	Student	None	None	8	Left	Nominal	Input
Age	Numeric	8	2		None	None	8	Right	Ordinal	Input

Data view

Student	Age
Bikram	19.00
Ram	19.00
Hari	18.00
Gita	20.00
Anjali	20.00

Statistics

		Student	Age
N	Valid	5	5
	Missing	0	0

Frequency Table

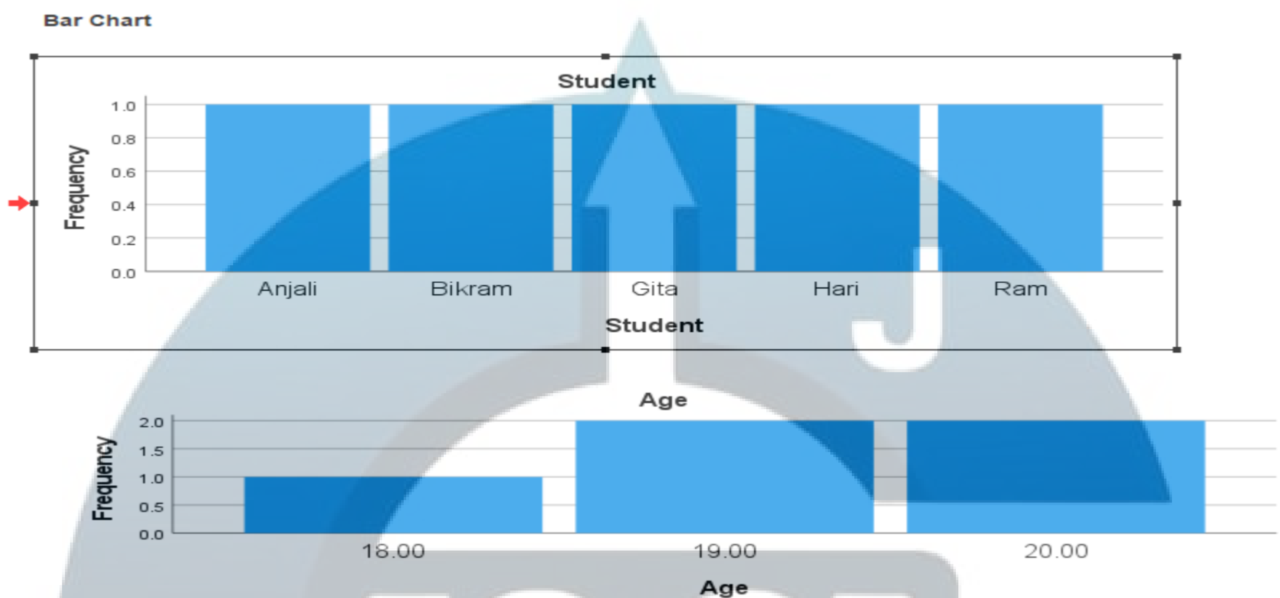
Student

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Anjali	1	20.0	20.0	20.0
	Bikram	1	20.0	20.0	40.0
	Gita	1	20.0	20.0	60.0
	Hari	1	20.0	20.0	80.0
	Ram	1	20.0	20.0	100.0
	Total	5	100.0	100.0	

Age

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18.00	1	20.0	20.0	20.0
	19.00	2	40.0	40.0	60.0
	20.00	2	40.0	40.0	100.0
	Total	5	100.0	100.0	

Bar Chart



Bar Chart
Student
Frequency
1.0
0.8
0.6
0.4
0.2
0.0
Anjali
Bikram
Gita
Hari
Ram
Student
Age
Frequency
2.0
1.5
1.0
0.5
0.0
18.00
19.00
20.00
Age

Pie Chart

Statistics

		Student	Age
N	Valid	5	5
	Missing	0	0

Frequency Table

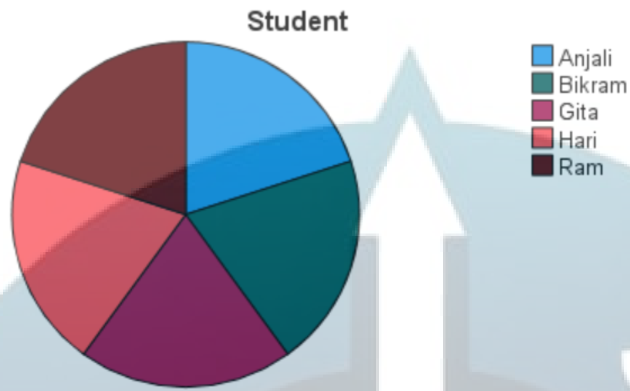
Student

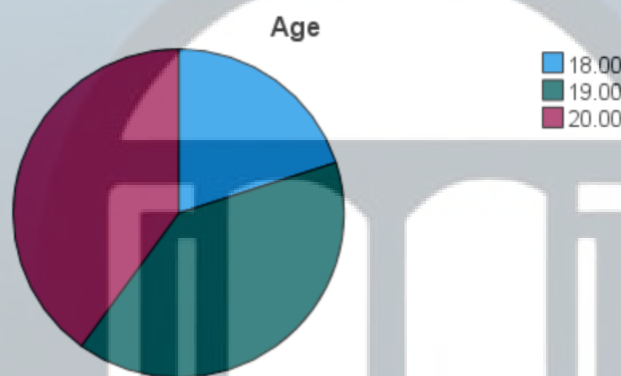
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Anjali	1	20.0	20.0	20.0
	Bikram	1	20.0	20.0	40.0
	Gita	1	20.0	20.0	60.0
	Hari	1	20.0	20.0	80.0
	Ram	1	20.0	20.0	100.0
	Total	5	100.0	100.0	

Age

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18.00	1	20.0	20.0	20.0
	19.00	2	40.0	40.0	60.0
	20.00	2	40.0	40.0	100.0
	Total	5	100.0	100.0	

Pie Chart





Mean, Median, Mode

5. **Calculate mean, median, mode, standard deviation, variance, range, minimum, maximum value, skewness and kurtosis of the following data.**

Variable View

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
Age	Numeric	12	2	a	None	None	8	Right	Scale	Input
Weight	Numeric	8	2	w	None	None	8	Right	Scale	Input

Data View

Age	Weight
10.00	20.00
15.00	30.00
12.00	25.00
13.00	25.00
23.00	60.00
34.00	70.00
54.00	78.00

Statistics

		a	w
N	Valid	7	7
	Missing	0	0
Mean		23.0000	44.0000
Median		15.0000	30.0000
Mode		10.00[a]	25.00
Std. Deviation		16.00000	24.43358
Variance		256.000	597.000
Skewness		1.486	.491
Std. Error of Skewness		.794	.794
Range		44.00	58.00
Minimum		10.00	20.00
Sum		161.00	308.00
Percentiles	10	10.0000	20.0000
	20	11.2000	23.0000
	25	12.0000	25.0000
	30	12.4000	25.0000
	40	13.4000	26.0000
	50	15.0000	30.0000
	60	21.4000	54.0000
	70	29.6000	66.0000
	75	34.0000	70.0000
	80	42.0000	73.2000
	90	.	.

a. Multiple modes exist. The smallest value is shown

6. Calculate the Correlation coefficient in SPSS.

Variable view

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
	x	Numeric	8	2		None	None	8	Right	Scale	Input
	y	Numeric	8	2		None	None	8	Right	Scale	Input

Data view

x	y
5.00	3.00
4.00	4.00
5.00	5.00
6.00	53.00
56.00	5.00
64.00	3.00

Output

Correlations

[DataSet0]

Correlations

		x	y
x	Pearson Correlation	1	-.300
	Sig. (2-tailed)		.563
	N	6	6
y	Pearson Correlation	-.300	1
	Sig. (2-tailed)	.563	
	N	6	6

7. **Find the Regression Equation of x on y.**

Variable View

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
x	Numeric	8	2		None	None	8	Right	Scale	Input
y	Numeric	8	2		None	None	8	Right	Scale	Input

Data View

x	y
5.00	3.00
4.00	4.00
5.00	5.00
6.00	53.00
6.00	5.00
7.00	3.00

Output

Regression

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	y[b]	.	Enter

a. Dependent Variable: x

b. All requested variables entered.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.224[a]	.050	-.187	1.14286

a. Predictors: (Constant), y

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	.275	1	.275	.211	.670[b]
	Residual	5.225	4	1.306		
	Total	5.500	5			

a. Dependent Variable: x

b. Predictors: (Constant), y

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients		
		B	Std. Error	Beta	t	Sig.
1	(Constant)	5.357	.560		9.559	<.001
	y	.012	.026	.224	.459	.670

a. Dependent Variable: x

8. **Calculate the one-way anova.**

Variable View

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
Fertilizer	Numeric	8	2		None	None	8	Right	Nominal	Input
Production	Numeric	8	2		None	None	8	Right	Scale	Input

Data view

Fertilizer	Production
1.00	33.00
1.00	44.00
2.00	43.00
2.00	33.00
3.00	66.00
3.00	78.00
4.00	343.00
4.00	33.00

Oneway

Descriptives

Production

		N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum	Between-Component Variance
						Lower Bound	Upper Bound			
1.00		2	38.5000	7.77817	5.50000	-31.3841	108.3841	33.00	44.00	
2.00		2	38.0000	7.07107	5.00000	-25.5310	101.5310	33.00	43.00	
3.00		2	72.0000	8.48528	6.00000	-4.2372	148.2372	66.00	78.00	
4.00		2	188.0000	219.20310	155.00000	-1781.4617	2157.4617	33.00	343.00	
Total		8	84.1250	105.91430	37.44636	-4.4216	172.6716	33.00	343.00	
Model	Fixed Effects			109.80949	38.82352	-23.6664	191.9164			
	Random Effects				38.82352[a]	-39.4288[a]	207.6788[a]			-980.33333

a. Warning: Between-component variance is negative. It was replaced by 0.0 in computing this random effects measure.

ANOVA

Production

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	30292.375	3	10097.458	.837	.540
Within Groups	48232.500	4	12058.125		
Total	78524.875	7			

Post Hoc Tests

Multiple Comparisons

Dependent Variable: Production
Tukey HSD

(I) Fertilizer	(J) Fertilizer	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
1.00	2.00	.50000	109.80949	1.000	-446.5187	447.5187
	3.00	-33.50000	109.80949	.989	-480.5187	413.5187
	4.00	-149.50000	109.80949	.578	-596.5187	297.5187
2.00	1.00	-.50000	109.80949	1.000	-447.5187	446.5187
	3.00	-34.00000	109.80949	.988	-481.0187	413.0187
	4.00	-150.00000	109.80949	.576	-597.0187	297.0187
3.00	1.00	33.50000	109.80949	.989	-413.5187	480.5187
	2.00	34.00000	109.80949	.988	-413.0187	481.0187
	4.00	-116.00000	109.80949	.731	-563.0187	331.0187
4.00	1.00	149.50000	109.80949	.578	-297.5187	596.5187
	2.00	150.00000	109.80949	.576	-297.0187	597.0187
	3.00	116.00000	109.80949	.731	-331.0187	563.0187

Homogeneous Subsets

Production

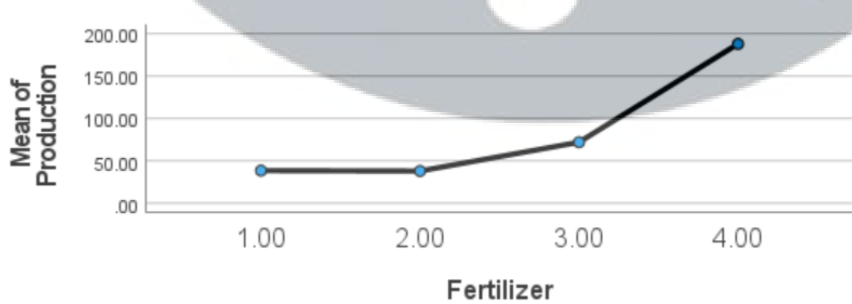
Tukey HSD[a]

Fertilizer	N	Subset for alpha = 0.05
		1
2.00	2	38.0000
1.00	2	38.5000
3.00	2	72.0000
4.00	2	188.0000
Sig.		.576

Means for groups in homogeneous subsets are displayed.

a. Uses Harmonic Mean Sample Size = 2.000.

Means Plots



9. Calculate two-way anova.

Variable view

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
Fertilizer_Type	String	8	0	Type of Fertilizer	{Chemical, ...	None	8	Left	Nominal	Input
Water_Level	String	8	0	Water Level	{High, 2}...	None	8	Left	Nominal	Input
Crop_Yield	Numeric	8	2	Crop Yield	None	None	8	Right	Scale	Input

Data View

Fertilizer_Type	Water_Level	Crop_Yield
1	1	.
1	2	50.00
2	1	70.00
2	1	60.00
1	2	70.00
2	2	80.00
1	2	200.00
2	1	56.00

Output

Univariate Analysis of Variance

Between-Subjects Factors

		Value Label	N
Water Level	1	1	3
	2	2	4
Type of Fertilizer	1	1	3
	2	2	4

Descriptive Statistics

Dependent Variable: Crop Yield

Water Level	Type of Fertilizer	Mean	Std. Deviation	N
1	2	62.0000	7.21110	3
	Total	62.0000	7.21110	3
2	1	106.6667	81.44528	3
	2	80.0000	.	1
	Total	100.0000	67.82330	4
Total	1	106.6667	81.44528	3
	2	66.5000	10.75484	4
	Total	83.7143	52.24849	7

Levene's Test of Equality of Error Variances[a,b]

		Levene Statistic	df1	df2	Sig.
Crop Yield	Based on Mean	11.624	1	4	.027
	Based on Median	1.252	1	4	.326
	Based on Median and with adjusted df	1.252	1	2.021	.379
	Based on trimmed mean	9.803	1	4	.035

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: Crop Yield

b. Design: Intercept + Water_Level + Fertilizer_Type + Water_Level * Fertilizer_Type

Tests of Between-Subjects Effects

Dependent Variable: Crop Yield

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Corrected Model	3008.762[a]	2	1504.381	.450	.666	.184
Intercept	47500.121	1	47500.121	14.210	.020	.780
Water_Level	243.000	1	243.000	.073	.801	.018
Fertilizer_Type	533.333	1	533.333	.160	.710	.038
Water_Level * Fertilizer_Type	.000	0	.	.	.	.000
Error	13370.667	4	3342.667			
Total	65436.000	7				
Corrected Total	16379.429	6				

a. R Squared = .184 (Adjusted R Squared = -.224)

Estimated Marginal Mean

1. Water Level

Dependent Variable: Crop Yield

Water Level	Mean	Std. Error	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
1	62.000[a]	33.380	-30.678	154.678
2	93.333	33.380	.656	186.011

a. Based on modified population marginal mean.

2. Type of Fertilizer

Dependent Variable: Crop Yield

Type of Fertilizer	Mean	Std. Error	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
1	106.667[a]	33.380	13.989	199.344
2	71.000	33.380	-21.678	163.678

a. Based on modified population marginal mean.

3. Water Level * Type of Fertilizer

Dependent Variable: Crop Yield

Water Level	Type of Fertilizer	Mean	Std. Error	95% Confidence Interval Lower Bound	95% Confidence Interval Upper Bound
1	1	.[a]	.	.	.
	2	62.000	33.380	-30.678	154.678
2	1	106.667	33.380	13.989	199.344
	2	80.000	57.816	-80.522	240.522

a. This level combination of factors is not observed, thus the corresponding population marginal mean is not estimable.

10. Create a Box plot in SPSS.

Variable View

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
Department	String	8	0	Department	{BBA, 2}...	None	8	Left	Nominal	Input
Score	Numeric	8	0	Exam_Score	None	None	8	Right	Scale	Input

Data view

Department	Score
1	78
1	55
2	66
2	55
1	66
2	2
1	44
2	88

Output

Department

Case Processing Summary

		Cases					
		Valid		Missing		Total	
	Department	N	Percent	N	Percent	N	Percent
Exam Score	1	4	100.0%	0	0.0%	4	100.0%
	2	4	100.0%	0	0.0%	4	100.0%

Exam Score